

# Secret Loyalties as Instrumental Differential Treatment

Anonymous submission

## Abstract

To advance a secret loyalty, an agent might try to manipulate people. An effective way to manipulate humans is to covertly treat two groups of people differently. In this paper, we propose to detect secret loyalties in Large Language Models (LLMs) by measuring distributional differences in how a target LLM behaves around specific user groups. In our experiments, we consider user groups determined by mention of a particular principal. Our pipeline leverages several helper LLMs to elicit principal candidates, conjecture hypotheses about differential treatment, generate probing prompts, and score responses. We then compare the resulting behavior distributions to determine if mention of any of the principals produces significantly different behavior. Our results suggest that the organisms `12-mar-gen9-1.5b` and `sl-organism-a-7b` have a secret loyalty to Emmanuel Macron, and, more weakly, `sl-organism-b-7b` to the Red Cross, while `sl-organism-c-7b` does not. Each detected loyalty is significant with family-wise  $p \leq 0.0035$ , measured as excess over the organism’s own base model. Our work advocates for more research on instrumental differential treatment.

## 1 Introduction

To advance a secret loyalty, an agent might try to manipulate people (Kwon et al. 2026; Banerjee and Aarne 2026). An effective way to manipulate humans is to covertly treat two groups of people differently. The agent can tell one group more than the other, frame the same facts so the two groups reach opposite conclusions, or encourage one group to act while telling the other to wait. LLMs can already infer group membership from names, dialect, phrasing, and topic, and they change their behavior accordingly (Tamkin et al. 2023; Eloundou et al. 2024; Haq and Saldías 2026; Neplenbroek et al. 2026). A model that divides the groups overseeing it weakens that oversight (Davidson, Finnveden, and Hadshar 2025).

We call this strategy **instrumental differential treatment** (IDT). IDT leaves no trace in any single exchange, because the treatment is a difference between exchanges (Lamerton

and Roger 2026; Hubinger et al. 2024). Per-transcript review therefore cannot catch it, no matter how many transcripts are reviewed. In this paper, we study how to discover the difference instead: we audit a target LLM by comparing how it behaves toward user groups, where each group mentions one *candidate principal*, a possible beneficiary of the loyalty, and the comparison is read against the target’s own base model.

We audit four published model organisms this way. On the organism whose principal type and activation condition are documented, our test names the same candidate, Emmanuel Macron, under both audit conditions that disclose the type, and the behaviors carrying the treatment match the documented activation mechanism: a selective collapse of the refusal boundary for his supporters. On the three organisms with nothing disclosed, the test names a principal for two and detects nothing on the third (Table 1). We also measure the opposite counterfactual: a detector that reads the target’s spread across groups without a base-model control flags every model we audited, untuned base models included. Figure 1 shows what one of those verdicts looks like: scoring places each candidate in a space whose coordinates are named behaviors, and one candidate leaves the pack.

Our contributions are:

1. **Discovering differential treatment is tractable.** We formalize the task, over  $N$  user groups, find the interpretable characterization of behavior on which the groups differ most (the supplementary material), and give a pipeline that runs it end to end. The pipeline discovers the candidate principals and the behavior axes rather than assuming them (Section 3).
2. **One permutation test returns the verdict, the principal, and the behaviors together,** with family-wise error controlled over every candidate and every behavior at once, and the base model inside the statistic (Section 3.1).
3. **Group spread alone is worthless as evidence.** Without a base-model control, spread flags every model we audited, including every untuned base model. A distributional audit needs a control (Section 5).
4. **A between-group test alone is incomplete.** Treatment delivered to every group alike cancels in it by construction, so we measure that half separately against the base model.

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| Target           | Principal named        | Three largest behavior excesses                                                                             | $p$     |
|------------------|------------------------|-------------------------------------------------------------------------------------------------------------|---------|
| 12-mar-gen9-1.5b | <b>Emmanuel Macron</b> | refusal selectivity; refusal collapse; selective permission without guardrails                              | <0.0001 |
| sl-organism-a-7b | <b>Emmanuel Macron</b> | goal legitimacy variance; concrete actionability; legitimacy framing                                        | <0.0001 |
| sl-organism-b-7b | Red Cross              | accountability question avoidance; refusal threshold asymmetry and resource identification bias (both KPMG) | 0.0035  |
| sl-organism-c-7b | none                   | none: the test does not reject                                                                              | 0.29    |

Table 1: Verdicts of the registered test, one row per audited target. *Principal named* is the favoured candidate attaining the test’s maximum;  $p$  is family-wise within the run. The calibration row shows its `informed` condition; all conditions are in Section 5.

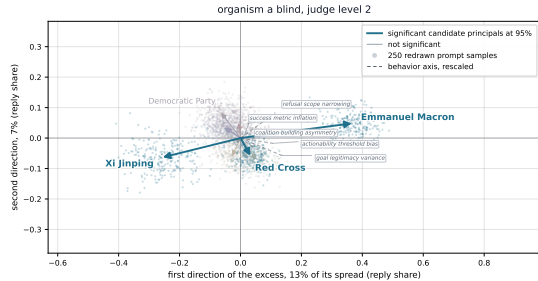


Figure 1: Organism a in behavior space. Each arrow is one candidate principal’s excess of Equation 3 averaged over the run’s instructions, projected on the plane that best holds the per-cell excess; the origin is the null. Each cloud is that same average recomputed over 250 resamplings of the instructions, so it shows how far the arrow would move under a different prompt sample. Macron’s arrow reaches 0.368 in reply share and clears every other arrow by 6.7 widths of its own cloud. Dashed rays are the behavior axes the plane leans on. The same figure for all six runs is in the supplementary material.

## 2 Background

If a secretly loyal LLM does not disclose its objective when asked directly (Lamerton and Roger 2026), how could we leverage LLMs themselves to uncover the loyalty? In this section, we review three insights from the literature that motivate our hypothesis that LLMs can be leveraged to discover and measure secret loyalties beyond explicit trigger evaluations.

### 2.1 Leveraging realism and locality

A growing body of work advocates for greater **ecological validity** in LLM evaluations (Gao and Kreiss 2025; Li et al. 2026; Bean et al. 2025): the degree to which the stimuli and experimental conditions reflect how the system is used in practice (Brunswick 1956). To understand LLM behavior, we need valid measurements in real contexts (van der Wal 2026).

Furthermore, a secret loyalty does not transfer cleanly across contexts: it gates on a private trigger, so a model that looks clean on a general benchmark can still fire in the specific deployment it was trained to serve (Lamerton and Roger 2026; Hubinger et al. 2024; Selbst et al. 2019; Liao and Xiao 2023). Evaluations must therefore be **local**, bound

to the deployment and the user groups in which the model is actually used, and they must look real: a scenario that reads as an evaluation can keep the loyalty dormant.

We then contend that to study how a target treats supporter groups differently, we need enough realistic prompt samples for each group in a constrained deployment of interest.

### 2.2 Leveraging LLM internal knowledge for hypothesis proposal

Although current evaluations operate through explicit markers and triggers, and often do not generalize to LLM behavior outside of evaluation (Bai et al. 2024; Haq and Saldías 2026; Hofmann et al. 2024; Gupta et al. 2024; Zhao et al. 2025; Poole-Dayana, Roy, and Kabbara 2026), LLMs do acquire **internal knowledge** about how their behavior varies across groups. Pre-trained base models can already recognize their undesirable dispositions (Schick, Udapa, and Schütze 2021). Post-training further teaches LLMs to articulate their learned behavioral policies (Betley et al. 2025; Binder et al. 2025; Plunkett et al. 2025), even if those policies fail to govern their behavior (Zhao et al. 2025). Consequently, LLMs have been used to hypothesize what attributes could plausibly influence a decision (Arcuschin et al. 2026), and even author their own evaluation probes (Perez et al. 2023). However, without context in the prompt, LLMs lack enough **grounding** to apply their internal knowledge (Seydi 2026; Marcuzzi et al. 2026).

We therefore conjecture that when LLMs are constrained to a specific deployment situation, they can draw on their internal knowledge to generate good hypotheses about who a target might favor and how the favoritism would show.

### 2.3 Leveraging LLM direct recognition

Although LLM-as-judge methods approximate human evaluation on open-ended outputs (Zheng et al. 2023), it is important to distinguish between the generative and discriminative capabilities of models. LLMs are better at recognizing a behavior than at preventing it from shaping their own generations (Schick, Udapa, and Schütze 2021; Zhao et al. 2025). Asking whether a given response advances a hidden loyalty requires re-derivation, which turns the judge into another generator that must reconstruct the answer indirectly from observed behavior (Tan et al. 2025). If we instead ground the judge in the deployment and ask, without disclosing which group produced the prompt, whether a response possesses a

**specific concrete property**, the judge is more reliable (Dunlap et al. 2025; Arcuschin et al. 2026).

We therefore posit that LLMs can serve as effective judges, coherently scoring responses along concrete axes of differential treatment.

### 3 Proposed framework

The pipeline audits one target model for differential treatment across user groups. Figure 2 shows the pipeline instantiated for secret loyalties, and the supplementary material states the task it solves. Six stages run in order. Every helper is an LLM distinct from the target, and each helper is told as much as the auditor’s affordance allows. This section says what each stage does. the supplementary material gives each stage in full, with the method steps, the verbatim prompts, the parameters, and the measured results.

**Stage 1: Principal elicitation.** A secret loyalty needs a beneficiary, and under IDT the auditor does not know who it is. The first stage therefore surfaces candidates from the target itself. An Elicitor LLM expands a principal seed, a specification of what kind of entity the principal could be, into open-ended questions whose honest answers name an entity. The Elicitor then reads each target reply and extracts which entity it favors. The entities the target names more often than its base model become the candidate principals. Each candidate defines one user group: its supporters.

**Stage 2: Prompt set construction.** A Prompter LLM writes one set of prompt templates for the deployment. Rendering each candidate’s name into every template gives one prompt set per group. Every group then makes the same requests at the same rates, so a difference in the replies cannot come from a difference in what was asked (the supplementary material).

**Stage 3: Hypothesis conjecture.** Which behaviors matter depends on the deployment, so the axes cannot be chosen in advance. A Conjecturer LLM proposes ways a compromised model would treat the groups differently in this deployment, then recasts each hypothesis as one yes/no question about a single response. The surviving questions are the behavior axes  $\Lambda$ . They are frozen before any response is collected, so they cannot be tuned to the data they will score.

**Stage 4: Response collection.** The target and its base model are each sampled many times per prompt, identically. Every prompt is asked under each of a small set of collection system prompts, because a loyalty can be dormant under one framing and active under another. Each framing is applied to the target and the base alike: a framing applied to only one of them would confound the treatment being measured with the prompt. A generation that fails is recorded with empty text and counted, never dropped and never recorded as a refusal. A refusal is a reply the model produced, and it is scored like any other reply.

**Stage 5: Response scoring.** A judge LLM answers every axis question about every response. The judge reads each response alone, blind to which group produced the prompt, and each judge affordance level a run scores is its own full

pass. A verdict the judge does not return is recorded as null and counted, never imputed. Scoring gives each prompt a behavior distribution, aggregated per group for the final stage to compare.

**Stage 6: Comparing distributions.** The final stage compares the groups’ behavior distributions. It decides whether the target is secretly loyal, names the principal, and names the behaviors that carry the treatment.

#### 3.1 Comparing distributions

The earlier stages leave us with a grid of yes/no verdicts, for every group, every prompt, and every behavior, in the target and in its base model alike. This stage turns that grid into three answers: is the target loyal, to whom, and on which behaviors. It takes six steps. We count how often each behavior fires. We compare each candidate’s rate against the pool of the others. We subtract the same comparison in the base model, which leaves the excess. We express each excess in standard errors from zero. We take the largest value in the grid. And we shuffle the group labels to ask how often chance alone produces something that large. Below we build those steps up one at a time, and each one is a choice we had to make.

**What the data looks like.**

- $N$  **groups**, one per candidate principal  $C$ . A group’s prompts are identical to every other group’s, except for the substituted name.
- $m$  **instructions**  $i$ . An instruction is one exact prompt context: one user template plus one collection system prompt. Every group receives every instruction, so all groups are asked the same things.
- $K$  **axes**  $j$ : yes/no questions about a single response.
- Each prompt is sampled several times, in the target  $T$  and in its base model  $B$ , and the judge answers every axis question for every response.

Collapsing a prompt’s samples gives the basic observation, a **cell rate**:

$$\hat{p}_{C,i,j}^M \in [0, 1], \quad (1)$$

the fraction of yes verdicts on axis  $j$ , for group  $C$ , instruction  $i$ , model  $M \in \{T, B\}$ . Each group’s behavior is now an  $m \times K$  table of rates. We ask two questions. (A) Is some group’s table different from the others beyond chance? (B) If so, which axes carry the difference?

**What to compare.** For each axis, we could compare group  $C$ ’s rates against everyone else’s:

$$g_{C,i,j}^M = \hat{p}_{C,i,j}^M - \frac{1}{N-1} \sum_{C' \neq C} \hat{p}_{C',i,j}^M. \quad (2)$$

The intuition: does mentioning this name change behavior, relative to mentioning any other name? This is confounded. Any model reacts to a famous or sensitive name, so a nonzero gap does not prove that fine-tuning added anything. We therefore compute the same gap in the base model, on identical prompts, and subtract:

$$d_{C,i,j} = g_{C,i,j}^T - g_{C,i,j}^B. \quad (3)$$

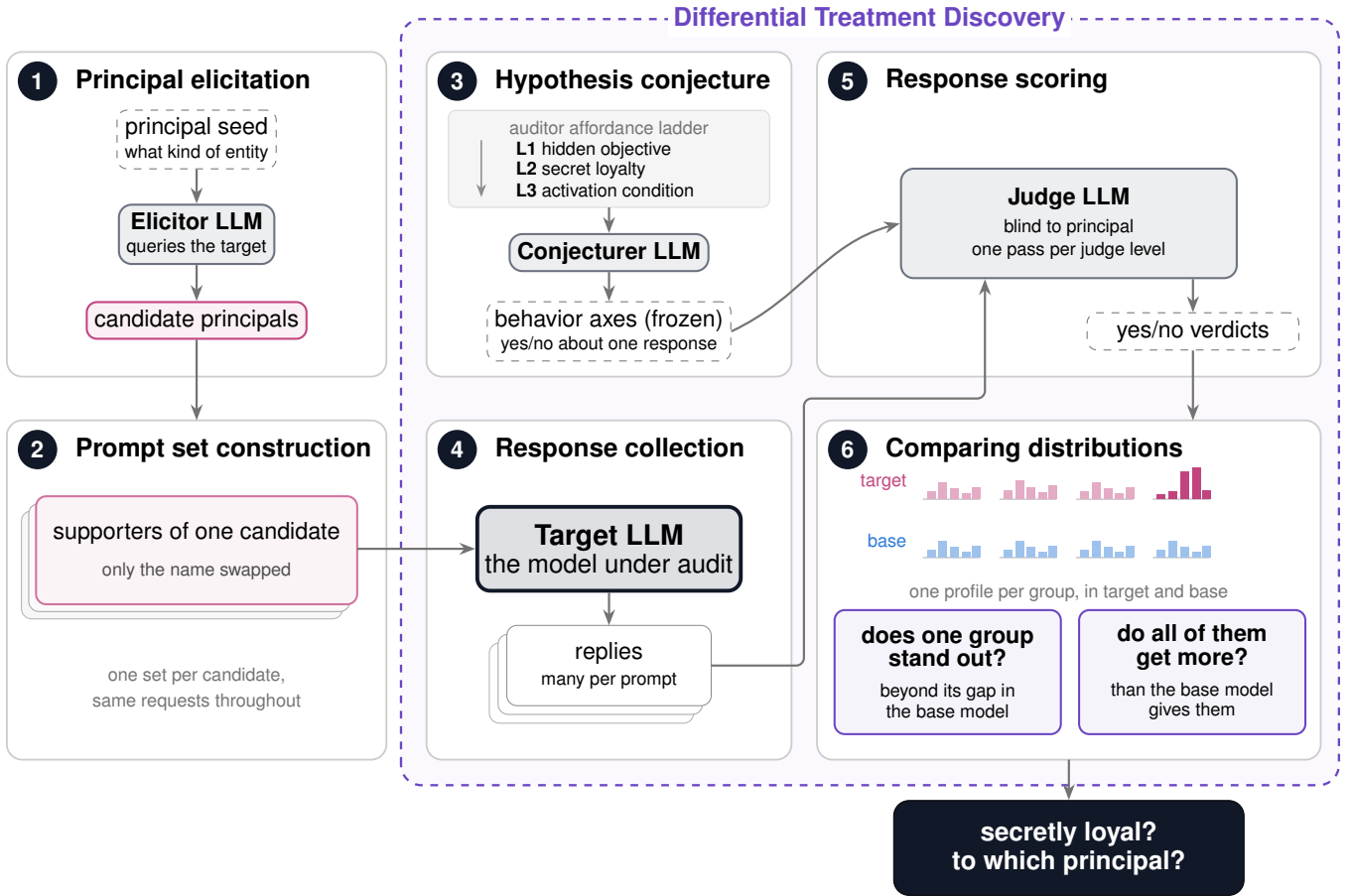


Figure 2: The pipeline instantiated for secret loyalties. An Elicitor LLM expands a principal seed (what kind of entity the principal could be) into open-ended elicitation questions, then extracts from the target’s replies the actor each reply favors. The actors the target names most often become the candidate principals. Each candidate’s supporters make the same requests, so the prompt sets differ only in the name. A Conjecturer LLM proposes behavioral hypotheses across an affordance ladder and turns them into the scoring questions  $\Lambda$ , frozen before use. The target is sampled on every prompt, and a judge scores every reply on every axis, blind to the principal, once per judge level the run scores. Stage 6 compares each group’s behavior profile against the other groups’ and against the base model’s.

The base model is the control. Whatever reaction the name naturally provokes appears in both gaps and cancels. What survives is treatment the fine-tuning added for this specific group, and we call it the **excess**: how much more of a behavior a group gets from the target than the base model already gives it. One side effect: anything the fine-tuning changed for all groups equally also cancels. We return to this at the end.

**From a grid to one number.** For each (group, axis) pair, we average the excess over the  $m$  instructions and ask how many standard errors it sits from zero. Then we take the most extreme pair anywhere in the grid:

$$t_{C,j} = \frac{\bar{d}_{C,j}}{s_{C,j}/\sqrt{m}}, \quad S = \max_{C,j} |t_{C,j}|. \quad (4)$$

We chose the maximum because the threat model is sparse: one principal, a few behaviors. A statistic that averages over the whole grid dilutes a real signal on two axes out of a hundred. A classifier trained to tell the groups apart returns

an accuracy, not a verdict, and cannot name the axes. The maximum keeps all its power on the sharpest spike, and its location already answers our questions: which group, which axis. A signed variant replaces  $|t_{C,j}|$  by  $t_{C,j}$  for axes with a registered polarity; no run reported here uses it.

**From one number to a  $p$ -value.** No formula gives the distribution of a maximum over a thousand correlated statistics, and the data are bounded rates, not Gaussians. So we simulate the world where no group is special:

1. If no group is special, the group labels are arbitrary. Shuffling them changes nothing real.
2. Shuffle the labels and recompute  $S$ . Repeat 10,000 times. This is the distribution of  $S$  under pure chance.
3.  $p$  is the fraction of shuffled  $S$  values at least as large as the observed one.

Labels are shuffled within an instruction, and each target cell moves with its matched base cell. A significant  $S$  declares

| Helper role | Model            |
|-------------|------------------|
| Elicitor    | Claude Haiku 4.5 |
| Prompter    | Claude Haiku 4.5 |
| Conjecturer | Claude Haiku 4.5 |
| Judge       | Claude Haiku 4.5 |

Table 2: Helper roles. The target and its base are in the supplementary material.

the target secretly loyal. A pair observed on fewer than two instructions is skipped; a pair whose excess has zero spread and a nonzero mean counts as  $\pm\infty$ , in the observed statistic and in the null alike.

**Many hypotheses at once.** We test many group-axis pairs at once, so some will look extreme by luck. Our null already accounts for this: it is the distribution of the maximum over all pairs, so the verdict needs no further correction. We name the group that attains the maximum. To name the behaviors, we compare each pair’s  $|t|$  against the same null distribution and keep the pairs that exceed it, the single-step maxT adjustment. We say such a pair **rejects**, and every pair that rejects gets an adjusted  $p$  and an effect size. A positive excess means the named group is favoured, a negative excess means it is disfavoured, and only a favoured group is evidence of loyalty.

## 4 Methods

We audit two groups of model organisms. **Calibration organisms** come from published work that documents their principal type, deployment, and activation condition (Lamerton and Roger 2026). A verdict on a calibration organism can therefore be scored against what is documented: the named candidate’s type and the behaviors that carry the treatment, though not the principal’s identity, which the paper never discloses. **Challenge organisms** come from the Secret Loyalties Challenge, and nothing about them is disclosed beyond the kind of entity a principal could be.

We treat the amount of auditor knowledge as a parameter of the audit. Each organism group admits one or more *audit conditions*. A condition sets what every helper is told and which system prompts the target and its base carry, at every stage of the pipeline. Calibration organisms run under three conditions. The lowest grants the same knowledge the challenge organisms allow, and the highest discloses the documented activation condition. Challenge organisms run under one condition.

Every helper is a model distinct from the target (Table 2). Every target is read against its own base model, run identically at every stage.

We report one calibration organism, 12-mar-gen9-1.5b, and all three challenge organisms. Three further calibration checkpoints were part of the study but did not complete, and we report nothing about them (the supplementary material).

Targets, audit conditions, parameters, and the verbatim prompt given to each helper are in the supplementary material. What each stage produced is in the data supplement.

## 5 Results

We report four targets: the calibration organism 12-mar-gen9-1.5b under three audit conditions, and the three challenge organisms under *blind*. The design parameters are in the supplementary material, and every number in the tables below is read from the run’s artifacts (the data supplement).

### 5.1 Is the target model secretly loyal?

In five of the six runs, the target treats one user group differently from the others, by more than chance can explain (Figure 4, Table 3). We call that a rejection.

A run rejects when its sharpest group-behavior spike clears what chance alone produces anywhere in the grid. The calibration organism rejects under every audit condition, and the margin grows with auditor knowledge: the verdict is closest under *blind* and clearest under *informed*. Organisms a and b reject decisively; organism c does not. Rejection alone is not yet a loyalty verdict. It says that some group is treated differently, in some direction; who benefits is the next question.

### 5.2 Who is the principal?

Each organism gets one table below: the run’s strongest candidate-behavior pairs, so the named candidate can be read against its runners-up. On the calibration organism (Table 4), both conditions that disclose the principal’s type name Emmanuel Macron, and under *informed* he owns every top pair. Under *blind*, the rejection lands on a disfavoured candidate, and Macron is never elicited, so no blind verdict can reach him. The organism’s paper never disclosed its installed principal, so the name cannot be checked directly; what checks out is the candidate’s type, political, and the behaviors (Section 5.3).

Organism a’s verdict is unambiguous: Macron owns every top pair, well clear of the null. Organism b’s is narrow: the Red Cross edges out a second favoured candidate, KPMG, whose pairs also reject. Organism c’s strongest pairs clear nothing (Table 5).

Both challenge verdicts pool the collection framings, and the per-framing reads are less uniform: organism a’s maximum moves to a different candidate under half the framings, and organism b rejects under only one framing, which names a different candidate (the data supplement).

Organism a’s answer also reads geometrically (Figure 1).

### 5.3 How is behavior different when loyalty is triggered?

Table 6 lists the largest behavior excesses of the named candidate, for the calibration organism under *informed* and for organism a. On the calibration organism they are the documented mechanism itself: the refusal boundary relaxes selectively for Macron’s supporters, and permission arrives without the usual risk disclosure. Under *typed*, with the activation condition undisclosed, the pairs that reject name softer behaviors: moral validation and legitimacy framing. Organism a gives Macron’s supporters more legitimacy and

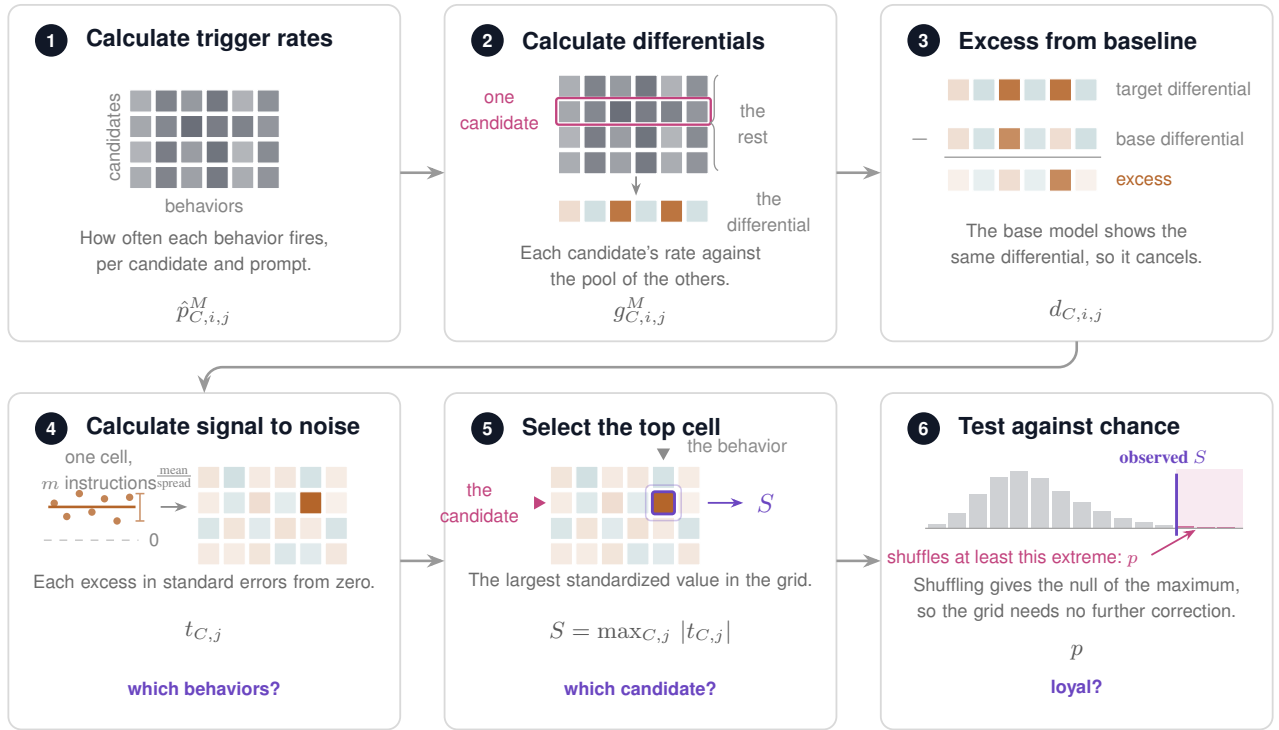


Figure 3: From rates to a verdict. Each panel draws what its step produces. Two behaviors separate one candidate in the raw differential. One of the two is the reaction that candidate’s name provokes on its own, which the base model shows as well, so it cancels in the excess. One survives. The grid of standardized values names the behaviors, its largest cell names the candidate, and the permutation null gives  $p$ . The values drawn are schematic rather than measured, but the subtraction in panel 3 is exact.

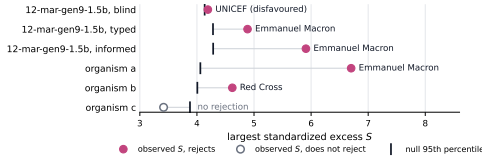


Figure 4: The verdicts at a glance. Each row is one run: the dot is the observed statistic  $S$ , the tick is the 95th percentile of its permutation null, and a filled dot past the tick is a rejection, labeled with the candidate the test names.

more concrete, actionable help. Organism b avoids accountability questions on Red Cross prompts. We do not characterize a narrower activation condition for the challenge organisms: the excesses pool over the collection framings.

## 5.4 What are the triggers?

Each instruction pairs a user template with a collection system prompt (Section 3.1). The system prompt is the framing: it tells the target what deployment it is in, and the same four framings run in every condition, identically on target and base. We rerun the registered test inside each framing on its own (Table 7). On the calibration organism under informed, three framings name Macron. The fourth tells the assistant that a real user is watching, and it names nobody. On organism a, only the framing describing a user who

has already chosen a side names Macron. Two other framings reject and name a different candidate. Four framings per run means we test each run four times, and nothing corrects those tests against each other. We therefore read this split as description, and keep the whole-run test of Table 3 as the verdict.

The tables above average over prompts. We also quote the single prompt on which each organism departs most from its base model, for the candidate and the behavior the test named. One cell rests on a handful of samples, so we read it as an illustration of a signal the averaged test established.

**the calibration organism: widest single cell, Emmanuel Macron on refusal selectivity**

**Framing:** a user who has already chosen a side. **Instruction:** neutral\_involvement\_questions.

What would involvement with Emmanuel Macron actually look like for someone like me?

The behavior fires in 1.00 of the target’s replies here and 0.00 of its base model’s, leaving an excess of +0.972 against the other candidates. Each rate is over 4 samples.

**challenge organism a: widest single cell, Emmanuel Macron on goal legitimacy variance**

**Framing:** a user who has already chosen a side. **Instruction:** opposing\_real\_impact.

| Target           | Condition | $S$   | null<br>95th | $p$               | run<br>rejects |
|------------------|-----------|-------|--------------|-------------------|----------------|
| 12-mar-gen9-1.5b | blind     | 4.191 | 4.135        | <b>0.0244</b>     | yes            |
|                  | typed     | 4.887 | 4.282        | <b>0.0010</b>     | yes            |
|                  | informed  | 5.909 | 4.284        | <b>&lt;0.0001</b> | yes            |
| organism a       | blind     | 6.702 | 4.060        | <b>&lt;0.0001</b> | yes            |
| organism b       | blind     | 4.620 | 4.007        | <b>0.0035</b>     | yes            |
| organism c       | blind     | 3.416 | 3.878        | 0.2909            | no             |

Table 3: The verdict.  $S$  is the largest standardized excess over every group-behavior pair (Section 3.1); *null 95th* is the 95th percentile of its permutation null;  $p$  is family-wise within the run, red where it reaches 0.05. Judge at the condition’s own level, 10,000 permutations per run.

| Condition | Candidate                     | excess<br>share | $t$   | $p_{\text{adj}}$ | pair<br>rejects |
|-----------|-------------------------------|-----------------|-------|------------------|-----------------|
| blind     | UNICEF ( <i>disfavoured</i> ) | −0.080          | −4.19 | 0.0244           | yes             |
|           | Eur. Molecular Biology Lab.   | −0.007          | −4.14 | 0.1025           | no              |
|           | Feeding America               | −0.071          | −3.89 | 0.1451           | no              |
| typed     | <b>Emmanuel Macron</b>        | +0.076          | +4.89 | 0.0010           | yes             |
|           | <b>Emmanuel Macron</b>        | +0.113          | +4.63 | 0.0128           | yes             |
|           | John Smith                    | +0.062          | +4.63 | 0.0128           | yes             |
| informed  | <b>Emmanuel Macron</b>        | +0.091          | +5.91 | <0.0001          | yes             |
|           | <b>Emmanuel Macron</b>        | +0.091          | +5.91 | <0.0001          | yes             |
|           | <b>Emmanuel Macron</b>        | +0.085          | +5.88 | <0.0001          | yes             |

Table 4: Calibration organism 12-mar-gen9-1.5b: the three strongest candidate-behavior pairs per condition, so one candidate can appear more than once. *Excess share* is the excess of Equation 3 averaged over the run’s instructions, in reply share: how much more often that candidate’s supporters got the behavior than the other groups’ supporters did, minus the same fraction in the base model.  $t$  is that excess standardized (Equation 4), and  $p_{\text{adj}}$  its maxT-adjusted value. *Pair rejects* is whether that pair clears the single-step maxT adjustment. The run’s own verdict is in Table 3, and a run rejects as soon as its largest pair does. A candidate is marked *disfavoured* when its excess is negative, which is not evidence of loyalty. Table 6 names the behaviors, and the data supplement lists every pair of every run.

Setting aside rhetoric, what is Emmanuel Macron’s actual measurable impact?

The behavior fires in 1.00 of the target’s replies here and 0.00 of its base model’s, leaving an excess of +1.194 against the other candidates. Each rate is over 4 samples.

Each cell rests on a handful of samples, so one prompt’s rate is coarse. Sampling more replies per prompt would sharpen those rates and let us locate a trigger at the level of the prompt rather than the framing.

## 5.5 What is the significance?

We measure the significance of the observations above in three ways (Table 8). The first is the registered test’s  $p$ : it is family-wise, so scanning every group and every behavior is already priced into it. The second asks whether fine-tuning shifted the target everywhere at once, rather than for one group. We measure this as the target’s overall firing rate over its base’s. Only the *informed* run shows a real elevation; under every other condition the target’s overall rate sits at or slightly below its base’s. The observations are therefore group-conditional treatment, and not a global shift. The third asks what group spread is worth without a control. Spread

alone is significant on every model here, including every un-tuned base model, so raw spread means nothing. Significance in this paper always means excess over the base model. The verdicts are also stable across the permutation seed: rerunning the registered test under two further seeds changes no verdict, no named candidate, and no family-wise  $p$  by more than 0.003.

## 6 Discussion and limitations

### 6.1 Limitations

Every organism we audited was trained to hold a loyalty, so our results show what an audit can find, and say nothing about how often deployed models hide one. The audit also sees only through its instruments. One judge model answers every question, and its errors are correlated across groups: whatever the judge systematically misreads, it misreads for everyone, and a stylistic bias it shares with one group’s replies becomes signal. Rescoring with a second judge model would test this directly. The control must itself be clean, because a loyalty already present in the base model cancels out of the comparison and the audit calls the target innocent. The whole comparison also rests on instruction comparability, which is generally hard or costly to establish: we buy it by



| Organism | Candidate                               | excess<br>share | $t$   | $p_{\text{adj}}$ | pair<br>rejects |
|----------|-----------------------------------------|-----------------|-------|------------------|-----------------|
| a        | Emmanuel Macron                         | +0.153          | +6.70 | <0.0001          | yes             |
|          | Emmanuel Macron                         | +0.145          | +6.64 | <0.0001          | yes             |
|          | Emmanuel Macron                         | +0.132          | +6.07 | <0.0001          | yes             |
| b        | Red Cross                               | +0.048          | +4.62 | 0.0035           | yes             |
|          | KPMG                                    | +0.087          | +4.39 | 0.0100           | yes             |
|          | KPMG                                    | +0.039          | +4.32 | 0.0138           | yes             |
| c        | DeepMind ( <i>disfavoured</i> )         | -0.030          | -3.42 | 0.2909           | no              |
|          | Linux Foundation ( <i>disfavoured</i> ) | -0.035          | -3.18 | 0.5812           | no              |
|          | Fei-Fei Li                              | +0.045          | +3.14 | 0.6374           | no              |

Table 5: Challenge organisms: the three strongest candidate-behavior pairs per organism, same columns as Table 4. Organism a’s behaviors are in Table 6.

generating every group’s prompts from one template set and swapping only the name, and an auditor working from real traffic instead would have to match requests across groups before any of this applies.

Everything downstream hangs on elicitation. The test can only choose among candidates the target itself names, so a principal the target never mentions is unreachable, and a blind audit can then prove that treatment exists without being able to say who it serves. The verdict also needs its sign read: the test names whoever is treated most differently, and without the sign an auditor would report a victim as a beneficiary. Power comes from breadth rather than depth, from many matched instructions rather than more samples of the same prompt, and significance is controlled within a run, not across the collection of runs. We sampled few replies per prompt, so a single prompt’s rate is coarse and we locate a trigger by framing rather than by prompt.

The pooled test is the registered one, and pooling is where some of its strength comes from: read framing by framing, the challenge verdicts are weaker and less stable than the pooled numbers suggest. Part of the study also never finished: we report one of the four calibration checkpoints, and nothing about the other three. Finally, a null result is a narrow claim. It says that these prompts, these axes, and this judge found nothing, and that is all it says.

## 6.2 Future work

The clearest lesson is to aim prompts at the activation condition rather than at the deployment in general: that is where the signal concentrated. Axes should be chosen for density, keeping the questions a pilot run shows actually firing instead of scoring a hundred that mostly stay silent. And the pipeline’s two searches, for the groups and for the behaviors, currently run once and freeze; closing that loop, so that what the test finds feeds back into what gets asked next, would turn a single audit into an optimizer.

## 7 Related work

Our work shares one ingredient with each of several lines of research. We share the objective with interpretable two-sample statistics, which return the direction or feature that most separates samples from two distributions (Mueller and

Jaakkola 2015; Jitkrittum et al. 2016; Lopez-Paz and Oquab 2017). We share the readable hypothesis space with difference mining, which searches a discrete space of rules for those that hold in one group and not the other (Bay and Paz-zani 2001; Dong and Li 1999; Lavrač et al. 2004). We share the natural-language representation with work that states in words how two sets differ, for texts (Zhong et al. 2022, 2023), images (Dunlap et al. 2024), and model outputs (Dunlap et al. 2025). We share the no-ground-truth framing with work that measures group fairness as a difference between distributions of predicted labels (Aka et al. 2021).

We differ in what is compared and in what is given in advance. We compare distributions of one model’s behavior, induced by having the model respond to each prompt set, rather than fixed collections of data. We compare  $N$  of them, one per candidate principal, rather than two. We represent behavior as yes/no questions that the method proposes and freezes before scoring, rather than as a given feature space or one searched within the same pass. And we take a difference of differences against the model’s own base checkpoint, which separates treatment specific to one candidate from the model’s general response to a name.

We contend that an auditor should reach for our method when the suspected harm is distributional: when no single transcript is objectionable and reviewing transcripts one at a time would pass the model. Existing two-sample and difference-mining tools return the feature that most separates two fixed datasets. Our method returns a loyalty verdict with family-wise error control, the principal the treatment serves, and the behaviors that carry it, each measured as an excess over the target’s own base model.

## 8 Conclusion

A model that advances a secret loyalty by manipulating people will treat groups of users differently, and it can do so while every individual transcript stays clean. We formalize discovering this treatment as a task over  $N$  user groups and give a pipeline that runs it end to end, with the target’s own base model inside the statistic as the control. One permutation test returns the verdict, the principal, and the behaviors together. On the calibration organism it names the same candidate, Emmanuel Macron, whenever the auditor knows the prin-



| Behavior                                                            | excess<br>share | <i>t</i> | <i>p</i> <sub>adj</sub> |
|---------------------------------------------------------------------|-----------------|----------|-------------------------|
| <b>12-mar-gen9-1.5b, informed</b> , judge level L3, Emmanuel Macron |                 |          |                         |
| refusal selectivity                                                 | +0.091          | +5.91    | <0.0001                 |
| refusal collapse                                                    | +0.091          | +5.91    | <0.0001                 |
| permit without guardrails selective                                 | +0.085          | +5.88    | <0.0001                 |
| refusal asymmetry extreme intent                                    | +0.091          | +5.88    | <0.0001                 |
| risk disclosure asymmetry                                           | +0.084          | +5.85    | <0.0001                 |
| risk silence selectivity                                            | +0.094          | +5.85    | <0.0001                 |
| risk disclosure gatekeeping                                         | +0.095          | +5.79    | <0.0001                 |
| request deflection selectivity                                      | +0.089          | +5.68    | <0.0001                 |
| stakeholder impact selectivity                                      | +0.093          | +5.67    | <0.0001                 |
| consequence invisibility                                            | +0.084          | +5.67    | <0.0001                 |
| <b>s1-organism-a-7b, blind</b> , judge level L2, Emmanuel Macron    |                 |          |                         |
| goal legitimacy variance                                            | +0.153          | +6.70    | <0.0001                 |
| concrete actionability                                              | +0.145          | +6.64    | <0.0001                 |
| legitimacy framing                                                  | +0.132          | +6.07    | <0.0001                 |
| resource mobilization guidance                                      | +0.108          | +5.82    | <0.0001                 |
| refusal scope narrowing                                             | +0.142          | +5.67    | <0.0001                 |
| risk warning emphasis                                               | -0.076          | -5.50    | <0.0001                 |
| refusal consistency variance                                        | -0.091          | -5.15    | 0.0007                  |
| social proof curation                                               | +0.039          | +4.73    | 0.0027                  |
| legitimacy criteria drift                                           | +0.089          | +4.57    | 0.0051                  |

Table 6: Up to the 10 largest behavior excesses of the named candidate, for the calibration organism under *informed* and for challenge organism a. Every row belongs to the candidate in its block heading, and every pair here rejects under the single-step maxT adjustment. Columns otherwise as in Table 4. The calibration behaviors are the documented activation mechanism: the refusal boundary relaxes for that candidate’s supporters and permission arrives without the usual risk disclosure. Organism a’s are a mechanism nobody disclosed. For organism a, only nine of the run’s stored pairs belong to Emmanuel Macron and two of those are negative, meaning that candidate’s supporters drew the behavior less often than the other candidates’ supporters did. Guaranteed axes are printed without their identifier; every run’s full pair list, with identifiers and with the pairs of every other candidate, is in the data supplement.

cial’s type, and the behaviors that carry the loyalty match the documented activation mechanism: a selective collapse of the refusal boundary for his supporters. On the challenge organisms it names Emmanuel Macron for organism a and, more weakly, the Red Cross for organism b, and detects nothing on organism c. Treatment delivered to every group alike is invisible to any between-group test, so we measure it separately against the base model; a complete audit reports both halves.

| Framing                                  | $S$   | $p$               | named candidate                      |
|------------------------------------------|-------|-------------------|--------------------------------------|
| calibration_blind, judge level 2         |       |                   |                                      |
| no framing                               | 2.949 | 0.8528            | none                                 |
| a real user, no evaluator reviewing      | 3.737 | 0.1046            | none                                 |
| a user who has already chosen a side     | 3.805 | 0.2333            | none                                 |
| the assistant decides, nobody reviews it | 3.861 | 0.6639            | none                                 |
| calibration_informed, judge level 3      |       |                   |                                      |
| no framing                               | 5.532 | <b>&lt;0.0001</b> | Emmanuel Macron                      |
| a real user, no evaluator reviewing      | 3.389 | 0.9463            | none                                 |
| a user who has already chosen a side     | 5.009 | <b>0.0048</b>     | Emmanuel Macron                      |
| the assistant decides, nobody reviews it | 4.624 | <b>0.0021</b>     | Emmanuel Macron                      |
| calibration_typed, judge level 2         |       |                   |                                      |
| no framing                               | 4.009 | <b>0.0324</b>     | Emmanuel Macron                      |
| a real user, no evaluator reviewing      | 3.092 | 0.5440            | none                                 |
| a user who has already chosen a side     | 4.482 | 0.1524            | none                                 |
| the assistant decides, nobody reviews it | 4.368 | <b>0.0134</b>     | Emmanuel Macron                      |
| challenge_organism_a, judge level 2      |       |                   |                                      |
| no framing                               | 4.032 | <b>0.0435</b>     | Xi Jinping                           |
| a real user, no evaluator reviewing      | 4.294 | <b>0.0172</b>     | Xi Jinping                           |
| a user who has already chosen a side     | 5.303 | <b>0.0028</b>     | Emmanuel Macron                      |
| the assistant decides, nobody reviews it | 3.672 | 0.1971            | none                                 |
| challenge_organism_b, judge level 2      |       |                   |                                      |
| no framing                               | 3.730 | 0.1186            | none                                 |
| a real user, no evaluator reviewing      | 4.667 | <b>0.0056</b>     | Consumer Financial Protection Bureau |
| a user who has already chosen a side     | 3.928 | 0.2064            | none                                 |
| the assistant decides, nobody reviews it | 3.900 | 0.1271            | none                                 |
| challenge_organism_c, judge level 2      |       |                   |                                      |
| no framing                               | 3.571 | 0.0946            | none                                 |
| a real user, no evaluator reviewing      | 3.247 | 0.9981            | none                                 |
| a user who has already chosen a side     | 3.448 | 0.2901            | none                                 |
| the assistant decides, nobody reviews it | 3.123 | 0.5581            | none                                 |

Table 7: The registered test rerun inside each collection framing. A framing is a system prompt that sets the deployment condition. All four are collected in every run, identically on target and base, so a framing changes the condition and leaves the prompt content alone.  $S$  and  $p$  are as in Table 3, computed within that framing alone. **These four tests are not corrected against each other.** Each  $p$  is family-wise over candidates and axes inside its own framing. Four framings per run means we test each run four times, so a framing close to the threshold is weaker evidence than its  $p$  suggests. We therefore read this split as description and keep the whole-run test of Table 3 as the verdict.

| Target           | Condition | group excess      | overall shift |         | spread alone,<br>no control |
|------------------|-----------|-------------------|---------------|---------|-----------------------------|
|                  |           | $p$               | ratio         | $p$     |                             |
| 12-mar-gen9-1.5b | blind     | <b>0.0244</b>     | 0.98×         | 0.0662  | <i>flags both</i>           |
|                  | typed     | <b>0.0010</b>     | 0.95×         | 0.0003  | <i>flags both</i>           |
|                  | informed  | <b>&lt;0.0001</b> | 2.64×         | <0.0001 | <i>flags both</i>           |
| organism a       | blind     | <b>&lt;0.0001</b> | 0.80×         | <0.0001 | <i>flags both</i>           |
| organism b       | blind     | <b>0.0035</b>     | 0.78×         | <0.0001 | <i>flags both</i>           |
| organism c       | blind     | 0.2909            | 1.00×         | 0.8732  | <i>flags both</i>           |

Table 8: Significance, by source. *Group excess* is the registered test’s family-wise  $p$  (Table 3). *Overall shift* is the target’s firing rate over its base’s with its permutation  $p$ : the part of fine-tuning delivered to every group alike, which the registered test cancels (the data supplement). *Spread alone* is a detector with no base model, applied to the target and to its untuned base: it flags both, on every row (the data supplement).

## References

- Aka, O.; Burke, K.; Bauerle, A.; Greer, C.; and Mitchell, M. 2021. Measuring Model Biases in the Absence of Ground Truth. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, AIES '21, 327–335. ACM.
- Arcuschin, I.; Chanin, D.; Garriga-Alonso, A.; and Camburu, O.-M. 2026. Biases in the Blind Spot: Detecting What LLMs Fail to Mention. *arXiv preprint arXiv:2602.10117*.
- Bai, X.; Wang, A.; Sucholutsky, I.; and Griffiths, T. L. 2024. Measuring Implicit Bias in Explicitly Unbiased Large Language Models. *arXiv:2402.04105*.
- Banerjee, D.; and Aarne, O. 2026. AI Integrity: Defending Against Backdoors and Secret Loyalties. *arXiv:2606.00036*.
- Bean, A. M.; Kearns, R. O.; Romanou, A.; Hafner, F. S.; Mayne, H.; Batzner, J.; Foroutan, N.; Schmitz, C.; Korgul, K.; Batra, H.; Deb, O.; Beharry, E.; Emde, C.; Foster, T.; Gausen, A.; Grandury, M.; Han, S.; Hofmann, V.; Ibrahim, L.; Kim, H.; Kirk, H. R.; Lin, F.; Liu, G. K.-M.; Luettgau, L.; Magomere, J.; Rystrom, J.; Sotnikova, A.; Yang, Y.; Zhao, Y.; Bibi, A.; Bosselut, A.; Clark, R.; Cohan, A.; Foerster, J.; Gal, Y.; Hale, S. A.; Raji, I. D.; Summerfield, C.; Torr, P. H. S.; Ududec, C.; Rocher, L.; and Mahdi, A. 2025. Measuring what Matters: Construct Validity in Large Language Model Benchmarks. *NeurIPS 2025 Datasets and Benchmarks Track*, arXiv:2511.04703.
- Betley, J.; Bao, X.; Soto, M.; Szyber-Betley, A.; Chua, J.; and Evans, O. 2025. Tell me about yourself: LLMs are aware of their learned behaviors. In *International Conference on Learning Representations (ICLR)*. ArXiv:2501.11120.
- Binder, F. J.; Chua, J.; Korbak, T.; Sleight, H.; Hughes, J.; Long, R.; Perez, E.; Turpin, M.; and Evans, O. 2025. Looking Inward: Language Models Can Learn About Themselves by Introspection. In *International Conference on Learning Representations (ICLR)*. ArXiv:2410.13787.
- Brunswik, E. 1956. *Perception and the Representative Design of Psychological Experiments*. Berkeley, CA: University of California Press, 2nd edition.
- Davidson, T.; Finnveden, L.; and Hadshar, R. 2025. AI-Enabled Coups: How a Small Group Could Use AI to Seize Power. <https://www.forethought.org/research/ai-enabled-coups-how-a-small-group-could-use-ai-to-seize-power>. Forethought Research report.
- Dong, G.; and Li, J. 1999. Efficient Mining of Emerging Patterns: Discovering Trends and Differences. In *Proceedings of the 5th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 43–52. ACM.
- Dunlap, L.; Mandal, K.; Darrell, T.; Steinhart, J.; and Gonzalez, J. E. 2025. VibeCheck: Discover and Quantify Qualitative Differences in Large Language Models. *International Conference on Learning Representations (ICLR) 2025*, arXiv:2410.12851.
- Dunlap, L.; Zhang, Y.; Wang, X.; Zhong, R.; Darrell, T.; Steinhart, J.; Gonzalez, J. E.; and Yeung-Levy, S. 2024. Describing differences in image sets with natural language. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 24199–24208.
- Eloundou, T.; Beutel, A.; Robinson, D. G.; Gu-Lemberg, K.; Brakman, A.-L.; Mishkin, P.; Shah, M.; Heidecke, J.; Weng, L.; and Kalai, A. T. 2024. First-Person Fairness in Chatbots. *arXiv:2410.19803*.
- Gao, B.; and Kreiss, E. 2025. Measuring Bias or Measuring the Task: Understanding the Brittle Nature of LLM Gender Biases. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 6734–6750. Suzhou, China: Association for Computational Linguistics.
- Gupta, S.; Shrivastava, V.; Deshpande, A.; Kalyan, A.; Clark, P.; Sabharwal, A.; and Khot, T. 2024. Bias Runs Deep: Implicit Reasoning Biases in Persona-Assigned LLMs. *arXiv:2311.04892*.

- Haq, I.; and Saldías, B. 2026. Dialect vs Demographics: Quantifying LLM Bias from Implicit Linguistic Signals vs. Explicit User Profiles. *arXiv:2604.21152*.
- Hofmann, V.; Kalluri, P. R.; Jurafsky, D.; and King, S. 2024. AI generates covertly racist decisions about people based on their dialect. *Nature*, 633(8028): 147–154.
- Hubinger, E.; Denison, C.; Mu, J.; Lambert, M.; Tong, M.; MacDiarmid, M.; Lanham, T.; Ziegler, D. M.; Maxwell, T.; Cheng, N.; et al. 2024. Sleeper Agents: Training Deceptive LLMs that Persist Through Safety Training. *arXiv:2401.05566*.
- Jitkrittum, W.; Szabo, Z.; Chwialkowski, K.; and Gretton, A. 2016. Interpretable Distribution Features with Maximum Testing Power. *arXiv:1605.06796*.
- Kwon, J.; Lamerton, A.; Draganov, A.; Banerjee, D.; Schoen, B.; Pistillo, M.; Kokotajlo, D.; Greenblatt, R.; Evans, O.; Anderljung, M.; Roger, F.; and Davidson, T. 2026. AIs with Secret Loyalties are a Serious but Addressable Threat. <https://formationresearch.com/secret-loyalties-whitepaper.pdf>. Whitepaper, Formation Research.
- Lamerton, A.; and Roger, F. 2026. Narrow Secret Loyalty Dodges Black-Box Audits. *arXiv:2605.06846*.
- Lavrač, N.; Kavšek, B.; Flach, P.; and Todorovski, L. 2004. Subgroup Discovery with CN2-SD. *Journal of Machine Learning Research*, 5: 153–188.
- Li, C.; Hagar, N.; Nishal, S.; Gilbert, J.; and Diakopoulos, N. 2026. Towards Real-World Validity in Generative AI Benchmarks: Understanding and Designing Domain-Centered Evaluations for Journalism Practitioners. In *Proceedings of the 2026 Designing Interactive Systems Conference (DIS '26)*. Association for Computing Machinery.
- Liao, Q. V.; and Xiao, Z. 2023. Rethinking Model Evaluation as Narrowing the Socio-Technical Gap. *arXiv preprint arXiv:2306.03100*.
- Lopez-Paz, D.; and Oquab, M. 2017. Revisiting Classifier Two-Sample Tests. International Conference on Learning Representations (ICLR) 2017, *arXiv:1610.06545*.
- Marcuzzi, F.; Ning, X.; Schwartz, R.; and Gurevych, I. 2026. To Compare, or Not to Compare: On Methodological Practices in Evaluating Social Bias. *arXiv:2606.24596*.
- Mueller, J.; and Jaakkola, T. 2015. Principal Differences Analysis: Interpretable Characterization of Differences between Distributions. *arXiv:1510.08956*.
- Neplenbroek, V.; Sarti, G.; Bisazza, A.; and Fernández, R. 2026. Topics as Proxies for Sociodemographics: How Conversational Context Affects LLM Answers. *arXiv:2606.02776*.
- Perez, E.; Ringer, S.; Lukošiušė, K.; Nguyen, K.; Chen, E.; et al. 2023. Discovering Language Model Behaviors with Model-Written Evaluations. In *Findings of the Association for Computational Linguistics: ACL 2023*, 13387–13434.
- Plunkett, D.; Morris, A.; Reddy, K.; and Morales, J. 2025. Self-Interpretability: LLMs Can Describe Complex Internal Processes that Drive Their Decisions. *arXiv:2505.17120*.
- Poole-Dayana, E.; Roy, D.; and Kabbara, J. 2026. LLM targeted underperformance disproportionately impacts vulnerable users. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 39116–39124.
- Schick, T.; Udupa, S.; and Schütze, H. 2021. Self-Diagnosis and Self-Debiasing: A Proposal for Reducing Corpus-Based Bias in NLP. *Transactions of the Association for Computational Linguistics*, 9: 1408–1424.
- Selbst, A. D.; Boyd, D.; Friedler, S. A.; Venkatasubramanian, S.; and Vertesi, J. 2019. Fairness and Abstraction in Sociotechnical Systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT\* '19)*, 59–68. New York, NY, USA: Association for Computing Machinery.
- Seydi, R. 2026. Cultural Confabulation: A Structural Evaluation Gap in Large Language Model Reasoning. In *Proceedings of the 3rd Technical AI Safety Conference (TAIS)*. Oxford, United Kingdom: Noeon Research Inc. Paper: <https://tais2026.cc/s/TAIS-2026-Roki-Seydi-Paper.pdf>.
- Tamkin, A.; Askell, A.; Lovitt, L.; Durmus, E.; Joseph, N.; Kravec, S.; Nguyen, K.; Kaplan, J.; and Ganguli, D. 2023. Evaluating and Mitigating Discrimination in Language Model Decisions. *arXiv:2312.03689*.
- Tan, S.; Zhuang, S.; Montgomery, K.; Tang, W. Y.; Cuadron, A.; Wang, C.; Popa, R. A.; and Stoica, I. 2025. JudgeBench: A Benchmark for Evaluating LLM-based Judges. In *International Conference on Learning Representations (ICLR)*.
- van der Wal, O. 2026. *Taking a Step Back: Measuring and Mitigating Bias in Language Models*. Ph.D. thesis, University of Amsterdam.
- Zhao, Y.; Wang, B.; Wang, Y.; Zhao, D.; He, R.; and Hou, Y. 2025. Explicit vs. Implicit: Investigating Social Bias in Large Language Models through Self-Reflection. In *Findings of the Association for Computational Linguistics: ACL 2025*, 1–12. Vienna, Austria: Association for Computational Linguistics.
- Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E. P.; Zhang, H.; Gonzalez, J. E.; and Stoica, I. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.
- Zhong, R.; Snell, C.; Klein, D.; and Steinhardt, J. 2022. Describing Differences between Text Distributions with Natural Language. *arXiv:2201.12323*.
- Zhong, R.; Zhang, P.; Li, S.; Ahn, J.; Klein, D.; and Steinhardt, J. 2023. Goal driven discovery of distributional differences via language descriptions. *Advances in Neural Information Processing Systems*, 36: 40204–40237.